DSSAT

Inputs → Processes → Outputs

DSSAT: Decision Support System for Agrotechnology Transfer

Prepared by/ Marwa Samy

What is DSSAT?

- Process-based crop simulation model
- Integrates weather, soil, crop, and management

DSSAT Conceptual Framework

INPUTS
Weather | Soil | Crop | Management

PROCESSES
Growth | Water | Nitrogen | Stress

OUTPUTS
Phenology | Biomass | Yield | ET
(daily and seasonally)

DSSAT Inputs	Weather data
	Soil profile
	Crop genetics
	Management practices
	Pests & diseases
	Experimental design

DSSAT

Weather Input File (.WTH)

- Variables:
  - Tmax, Tmin (°C)
  - Net Downward Solar Radiation (SRAD)
  - Rainfall (RAIN)
  - Optional: Wind speed, (WIND)
  - Humidity
- **Daily** time step
- Controls:
  - Phenological development
  - Evapotranspiration (ET)
  - Soil water balance

DSSAT

Soil Input File (.SOL)

Soil profile layers (Soil layers, cm "SLB")

- Texture and bulk density (SLTX, SBDM)
- Field capacity & wilting point (SDUL, SLLL ".SOL")
- Organic carbon (SLOC ".SOL")
- Initial water & nitrogen (SH20, SANI or SLNI ".SOL")

- Controls
  - root growth
  - water/nutrient availability

DSSAT

Plant Characteristics Inputs (.CUL)

Cultivar genetic coefficients (.CUL)

- Phenology
- Biomass accumulation
- Yield formation

Crop Management Inputs

- Planting date
- Plant population at seeding
- Irrigation dates and amounts
- Fertilizer dates, type, and rate

Pests and Diseases Inputs

- Insects
- Diseases
- Weeds
- Stress impact on growth and yield

Experimental Design Inputs File (.X)

- Central DSSAT control file
- Links:
  - Weather (.WTH)
  - Soil (.SOL)
  - Crop model (e.g., CERES-Wheat)
- Defines:
  - Date of seeding
  - Plant density
  - Irrigation scheduling
  - Fertilizer applications
  - Simulation period

DSSAT Simulation Processes

- Phenological development
- Photosynthesis
- Biomass accumulation
- Soil water balance
- Nitrogen dynamics (Nitrogen balance)
- Stress response

DSSAT Outputs

- Crop phenology
- LAI and plant height
- Biomass (leaf, stem, grain)
- Evapotranspiration
- Grain yield
- Stress indices

- **.OUT files** – Main output files for simulated crop growth and development.
- **.GRO or .SUM files** – Summarized results per season / experiment.
- **.LOG files** – Detailed simulation logs (less commonly used for phenology).

DSSAT

Crop Phenology Outputs

- Emergence
- Anthesis (Flowering)
- Grain filling
- Physiological maturity
- Leaf Area Index (LAI)

Water Balance & Evapotranspiration Outputs

- • Actual evapotranspiration (ET)
- • Soil water content by layer
- • Drainage and runoff
- • Water stress factor (0–1)

DSSAT

Biomass and Growth Outputs

- Above-ground biomass
- Leaf biomass
- Stem biomass
- Root biomass
- Grain biomass

Yield Outputs

- Grain yield (kg ha^{-1})
- Harvest index
- Grain number
- Grain weight

Applications of DSSAT

- Climate change impact assessment
- Irrigation scheduling
- Fertilizer optimization
- Yield forecasting and decision support

Thank you
Any Question